China's Chemical industry Value Chain & Relevant Commodity Market Strategies

From Crude Oil to Polyester - A Tillion-RMB Chain

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Objective

Focus: Layer 1 crude oil (INE SC) and aromatics pass-through

Outline

- Value Chain: Crude Oil to Polyester
- Market Characteristics: Structure & Participants
- Market Characteristics: Data Pattern
- Marcoeconomic Shock Regime Detection

Goal: Study how crude oil landed cost behave onshore in China and its inference to the polyester margins through the chain. The main focus will be placed on layer 1 of the chain — SC

Value Chain: Crude Oil to Polyester

As of 2026-06-23

Layer 1 anchor (crude)

- **SC** (INE): China landed-cost benchmark — CNY/bbl, physical delivery (listed 26 Mar 2018).
- China is an import reliant country: **72%** import dependence (2024 EIA); refinery CDU util **57.5%** (-15.8 pp vs Feb baseline).
- SC **backwardation** + low INE receipts ⇒ tight deliverable; SC $-$**2.7 USD/bbl** vs Brent ⇒ weak refining demand.

Upstream | **Midstream** | **Downstream**

Crude		Naphtha/PX		PTA		EG		Polyester
SC INE	refine →	PX spot	oxid. →	**TA** ZCE	mrg poly →	**EG** DCE	+TA →	spot/CCF
507 CNY/bbl		8,800 CNY/t		6,028 CNY/t		4,495 CNY/t		Zhejiang
11.1 Mb/d		PX−naph 556		OI 1.03M		OI 310k		fiber/bottle
−2.7 vs Brent		z=+2.8		basis −252		parity −442		no future

■ Futures (SC, TA, EG) ■ Spot only

Value Chain: Key Spreads & Pass-Through

Crude sets cost floor; PTA is the current pinch point

Spread	Level	z	Interpretation
SC−Brent	−2.7 USD/bbl	−1.0	China discount
PX−naphtha	556 USD/t	+2.8	Wide margin
PX−TA (PTA)	2,772 CNY/t	+0.5	Compressing
TA−EG	1,533 CNY/t	+1.0	Wide margin
EG import parity	−442 CNY/t	−1.0	Import arb open

PX−naphtha uses Brent-implied naphtha proxy; EG parity vs CFR China landed.

Layer synthesis

- **L1:** Cheap SC + run cuts ⇒ demand-side stress.
- **L2:** Wide PX−naphtha despite run cuts — integrators defend aromatics cut.
- **L3:** PX rose faster than TA (−1,144 CNY/t margin over 20d); TA basis −252 (z=−4.5).
- **L4:** EG below import parity; futures deeply below spot (z=−5.2).

Market Characteristics: Structure & Ticker

INE SC — China's crude oil futures benchmark

Contract specs (SC)

- Exchange: **INE** (Shanghai International Energy Exchange)
- Quote: **CNY/bbl**; size **1,000 bbl/lot**; tick 0.1 CNY
- Delivery: **physical** into INE-designated bonded coastal tanks
- Spec: medium sour — API 32°, sulfur 1.5% wt
- first mainland energy future open to overseas investors

Benchmark history

Period	Benchmark
Pre-1993	No active crude futures
1993–2017	Offshore Brent / Dubai hedges
Mar 2018–now	**INE SC** + Brent/Dubai offshore

Economic role

- proxies **China import landed cost** (FX + freight + INE microstructure)

Futures linkage downstream

- SC → naphtha (spot) → PX (spot) → **TA** (ZCE) + **EG** (DCE)
- No single exchange spread for full chain — trade legs separately

Market Characteristics: Market Participants

Hubs, exchanges, and listed exposure

Geographic hubs

Hub	Province	Assets	Market size
Dalian / Changxing	Liaoning	Hengli integrated	400 kbpd CDU
Zhoushan / Ningbo	Zhejiang	Rongsheng ZPC	800 kbpd CDU
Shaoxing / Xiaoshan	Zhejiang	PTA / polyester cluster	∼30 Mt/yr PTA
Shandong coast	Shandong	Teapot refiners	∼4.2 Mb/d CDU

CDU = crude distillation; Shandong teapots ≈ 25% of national refining capacity.

Layer 1 players (crude)

- **Majors:** PetroChina, Sinopec, CNOOC
- **Independents:** Shandong teapots — active SC participants; util. **43.5%** vs majors **67.2%**
- **Merchants:** Glencore, Trafigura, Mercuria (reported at SC launch)
- **Government run guidance** — majors and megaprojects (59.2% util.) hold up better than teapots during margin squeeze.

Market Characteristics: Data Pattern

Layer 1 signals — Evidence of Recent Tight Demand (Data are as of 2026-06-18)

A. SC futures curve (INE)

Front: **SC2608** — 510.5 CNY/bbl, OI 50,948
6M: **SC2702** — 508.0 CNY/bbl
spread: $-$**2.5 CNY/bbl** (backwardation)
(Ann) roll yield: $-$**1.0%** — negative carry

B. SC basis vs Brent (vs 3y)

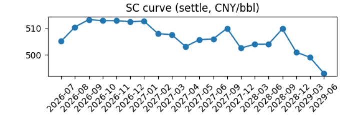
SC $\approx$ **75.0 USD/bbl** vs Brent **77.7**
Basis: $-$**2.7 USD/bbl** (-3.5%, $z = -1.0$)
Takeaway: China crude *cheap* vs offshore

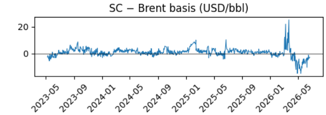
C. National crude balance (EIA 2024)

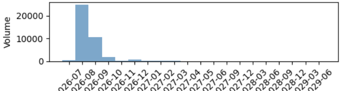
Imports **11.1 Mb/d** (-1.8% YoY)
Domestic production **4.3 Mb/d**
Refinery runs **14.2 Mb/d** (-4.1% YoY)
Import dependence: **72%**

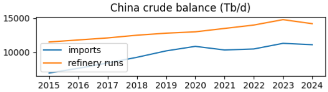
D. Refinery utilization (OilChem)

All-refinery CDU: **57.5%** (week ending)
vs pre-shock baseline (Feb 2026): $-$**15.8 pp**
Teapots **43.5%** | Majors **67.2%** |
Megaprojects **59.2%**





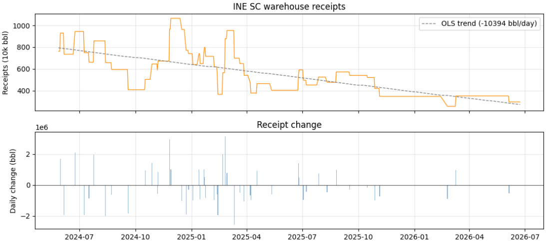




Market Characteristics: Data Pattern

E. INE warehouse receipts

- Latest: **2.96M bbl** (2026-06-23); flat 10+ sessions
- 20d change: $-$**550k bbl**; $z = -1.2$ vs sample
- Trend: drawn from ~10M to ~3M since 2024



Summary: China's marginal refining demand is weak, so SC trades at a discount to Brent — but near-term physical deliverable supply is still tight (backwardation, low warehouse receipts). Downstream, crude cost relief has not fully passed to PTA; EG looks domestically cheap vs imports.

Marcoeconomic Shock Regime Detection

Macro signal

- 2024–25: structural shocks **near zero** — no 2020 supply or 2022 fear episode
- Oil-specific realizations **faded**; price channel still precautionary structurally
- Global shocks quiet; China frictions dominate SC

Results

- SC move is **China margin / deliverable**, not global precautionary premium.
- Not all oil moves are the same.
- Supply doesn't implies price spike: supply shock moves production more than price.

Kilian BVAR model (data: 2018-03 – 2025, monthly):

$$z_t = c + \sum_{i=1}^{p} A_i z_{t-i} + e_t, \quad e_t = \begin{pmatrix} e_t^{\Delta prod} \\ e_t^{rea} \\ e_t^{rpo} \end{pmatrix} = \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} \epsilon_t^{\text{oil supply}} \\ \epsilon_t^{\text{aggregate demand}} \\ \epsilon_t^{\text{oil-specific demand}} \end{pmatrix} \quad (1)$$

where $z_t = (\Delta prod_t, rea_t, rpo_t)^T$, (rea) stands for real activity of global commodity

Future Work

- Investigate the reason of the weakening China refinery-demand.
- **Policy characteristics** of SC; How does the regulation differs from China to another other markets?
- **Layer 2 — Naphtha / PX:** Replace Brent-naphtha proxy with OilChem series; daily PX spot vs ZCE PX futures basis; link PX−naphtha to SC shocks and refinery CDU util.
- **Layer 3 — PTA (TA):** PTA margin (PX−TA) pass-through regressions from BVAR structural shocks; TA basis regime vs polyester operating rates (Zhejiang / Jiangsu).
- **Layer 4 — EG:** EG import parity vs CFR China (Wind); port inventory mean-reversion; TA−EG spread as tradable chain signal.
- **Layer 5 — Polyester:** CCF / SCI spot (fiber, bottle chip, film); demand-side falsifiers for upstream margins; complete executive chain summary.

Thank you for your time!

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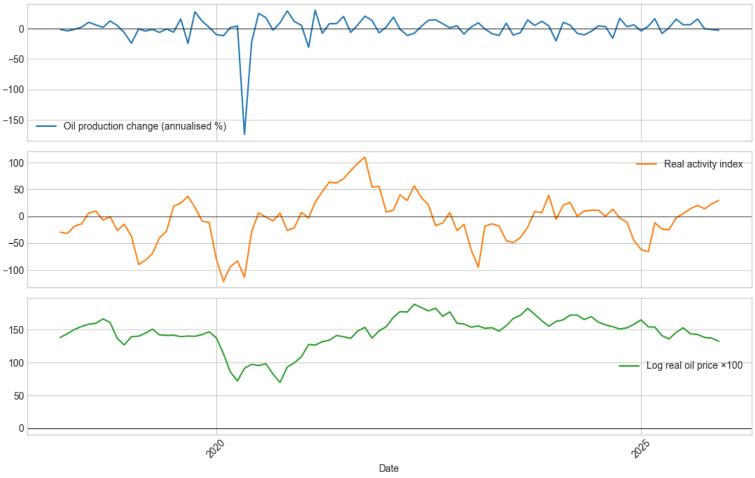
Appendix: Time Series Plot for BVAR model



Appendix: IRFs for BVAR model